1. `shows` Table

- **Normal Form Compliance: BCNF**
- **Why:** The primary key is `show_id`. All attributes (`movie_id`, `screen_id`, `start_time`, `end_time`, `status`, `created_at`) are functionally dependent *only* on the candidate key `show_id`. There are no partial dependencies (2NF) or transitive dependencies (3NF), and every determinant is a candidate key (BCNF).

show_id	movie_id	screen_id	start_time	end_time	status	created_at
101	501	1	2026-06-10 18:00:00	2026-06-10 21:00:00	ACTIVE	2026-06-01 10:00:00
102	502	2	2026-06-10 19:30:00	2026-06-10 22:15:00	ACTIVE	2026-06-01 10:05:00

2. `seats` Table (Static Layout)

- **Normal Form Compliance: BCNF**
- **Why:** The primary key is `seat_id`, and there is a composite candidate key `(screen_id, row_identifier, seat_number)`. All columns (`screen_id`, `row_identifier`, `seat_number`, `seat_category`) depend entirely on the keys. No non-prime attribute depends on another non-prime attribute.

seat_id	screen_id	row_identifier	seat_number	seat_category
1001	1	A	1	NORMAL
1002	1	A	2	NORMAL
1003	1	E	10	VIP

3. `show_seats` Table (Dynamic Locking State)

- **Normal Form Compliance: BCNF**
- **Why:** The composite primary key is `(show_id, seat_id)`. The status attributes (`status`, `locked_by_user_id`, `lock_expires_at`, `version`) describe the state of that *specific seat for that specific show*. They depend fully on the composite key.

show_id	seat_id	status	locked_by_user_id	lock_expires_at	version
101	1001	BOOKED	9821	NULL	2
101	1002	LOCKED	4512	2026-06-10 17:45:00	1
101	1003	AVAILABLE	NULL	NULL	1

4. `bookings` Table

- **Normal Form Compliance: BCNF**
- **Why:** Primary key is `booking_id`. Non-key attributes (`user_id`, `show_id`, `status`, `total_amount`, `expires_at`, etc.) depend entirely on `booking_id`. There are no hidden dependencies.

booking_id	user_id	show_id	status	total_amount	expires_at	created_at	updated_at
BKG-99211	4512	101	PENDING	550.00	2026-06-10 17:45:00	2026-06-10 17:35:00	2026-06-10 17:35:00
BKG-99212	9821	101	CONFIRMED	1100.00	2026-06-10 17:50:00	2026-06-10 17:40:00	2026-06-10 17:41:00